Phase 5 Routing Results (2026-05-14)

This note records the first completed routing-benchmark evidence pass after the phase-5 protocol expansion.

The routing phase is now evidence-complete in the narrow sense that the official replicated suites were run and aggregated for:

- symmetric TSPLIB95 Euclidean TSP
- asymmetric TSPLIB95 ATSP
- CVRPLIB capacitated vehicle routing

Each family used:

- 20 paired offsets
- 4 compared conditions
- paired bootstrap delta summaries on the main endpoints

The compared routing conditions were:

1. no replay
2. random replay
3. failure replay
4. failure replay plus compression pressure

Main Outcome

The current evidence is mixed and family-dependent.

- TSP: random_replay is the clearest winner.
- ATSP: failure_replay_compression is weakly favorable on combined transfer, but not cleanly on the primary held-out ATSP endpoint.
- CVRP: no_replay is the strongest overall baseline.

This means the phase does not currently support a broad claim that raw-gap failure replay is the dominant routing recipe across benchmark families.

Family Summaries

TSP

Best mean condition:

- tsplib_random_replay

Main paired result versus tsplib_no_replay:

- held-out TSPLIB gap delta: -0.0311, 95% CI [-0.0602, -0.0008]
- combined transfer-gap delta: -0.0282, 95% CI [-0.0520, -0.0029]

Interpretation:

- replay can help on a real routing family
- the strongest TSP gain came from random_replay, not failure_replay
- compression pressure sharply reduced accepted novelty relative to plain failure replay, but without a clean endpoint win

ATSP

Best mean condition:

- atsp_failure_replay_compression

Main paired result versus atsp_no_replay:

- combined transfer-gap delta: -0.0063, 95% CI [-0.0133, -0.0002]
- held-out ATSP-gap delta: -0.0056, 95% CI [-0.0191, 0.0071]

Interpretation:

- the compression-aware arm is directionally encouraging
- the primary held-out ATSP endpoint is not decisively separated
- this is a weakly favorable rather than strongly confirmatory result

CVRP

Best mean condition:

- cvrp_no_replay

Narrow paired result:

- cvrp_failure_replay beats cvrp_random_replay on held-out CVRPLIB gap with delta -0.0076, 95% CI [-0.0146, -0.0006]

But:

- cvrp_failure_replay does not beat cvrp_no_replay
- cvrp_failure_replay_compression lowers complexity relative to no_replay without improving the main endpoint

Interpretation:

- replay is not the dominant recipe on this family
- CVRP currently acts more as a boundary case than as a confirmation

Research Interpretation

The strongest defensible reading is:

- replay-aware machinery can matter on real routing benchmarks
- the effect is conditional on benchmark family
- the current routing results do not justify a general replay-dominance claim
- the current routing results also do not justify a strong "reusable heuristic compression" claim across all three families

This narrows the thesis claim from "replay always helps" to "replay pressure interacts with problem structure, and its benefits do not transfer uniformly across routing families."

Primary Sources

- TSP aggregate report
- ATSP aggregate report
- CVRP aggregate report
- TSP aggregate summary JSON
- ATSP aggregate summary JSON
- CVRP aggregate summary JSON